

1 Metric Spaces & General Topology

1.1 Metric Spaces, Topological Properties, and Completeness

We now abstract the topological concepts developed for R in Chapter 1 to arbitrary sets equipped with a notion of distance. This abstraction provides the theoretical foundation for functional analysis, where the points in the space are often themselves functions.

A **metric space** is a pair (X, d) , where X is a set and $d : X \times X \rightarrow R$ is a function (the metric) satisfying the following properties for all $x, y, z \in X$:

1. **Positivity:** $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$.
2. **Symmetry:** $d(x, y) = d(y, x)$.
3. **Triangle Inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.

The topology of a metric space is generated by open balls. For $x \in X$ and $r > 0$, the open ball of radius r centered at x is defined as $B_r(x) = \{y \in X \mid d(x, y) < r\}$. A subset $U \subset X$ is **open** if for every $x \in U$, there exists an $r > 0$ such that $B_r(x) \subset U$. A subset $F \subset X$ is **closed** if its complement $X \setminus F$ is open.

A sequence $(x_n)_{n=1}^{\infty}$ in a metric space (X, d) is a **Cauchy sequence** if for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $m, n \geq N$, $d(x_m, x_n) < \epsilon$. The metric space (X, d) is **complete** if every Cauchy sequence in X converges to a limit in X .

A particularly important class of complete metric spaces arises from vector spaces equipped with a norm.

A **Banach space** is a vector space V over R or C equipped with a norm $\|\cdot\|$ such that the metric space induced by the metric $d(u, v) = \|u - v\|$ is complete.

Completeness is a powerful property because it guarantees the existence of a limit based solely on the internal behavior of the sequence, without prior knowledge of the limit point. This principle is exploited in one of the most fundamental results in non-linear analysis: the Banach Fixed-Point Theorem.

Let (X, d) be a metric space. A map $T : X \rightarrow X$ is a **contraction mapping** if there exists a constant $c \in [0, 1)$ such that for all $x, y \in X$:

$$d(T(x), T(y)) \leq c d(x, y)$$

[Banach Fixed-Point Theorem] Let (X, d) be a non-empty complete metric space, and let $T : X \rightarrow X$ be a contraction mapping with contraction constant $c \in [0, 1)$. Then T admits a unique fixed point $x^* \in X$ (i.e., $T(x^*) = x^*$).

We first prove uniqueness. Suppose $x, y \in X$ are both fixed points of T , such that $T(x) = x$ and $T(y) = y$. Applying the contraction property yields:

$$d(x, y) = d(T(x), T(y)) \leq c d(x, y)$$

This implies $(1 - c)d(x, y) \leq 0$. Since $c < 1$, we have $1 - c > 0$. Because distance is non-negative ($d(x, y) \geq 0$), the only way this inequality can hold is if $d(x, y) = 0$. By the positivity axiom of a metric space, $x = y$. Thus, if a fixed point exists, it is strictly unique.

To prove existence, we construct a sequence via Picard iteration. Choose an arbitrary initial point $x_0 \in X$. Define the sequence $(x_n)_{n=0}^\infty$ recursively by $x_n = T(x_{n-1})$ for all $n \geq 1$. We must show that (x_n) is a Cauchy sequence.

Consider the distance between consecutive terms. By repeated application of the contraction property:

$$d(x_{n+1}, x_n) = d(T(x_n), T(x_{n-1})) \leq c d(x_n, x_{n-1}) \leq \dots \leq c^n d(x_1, x_0)$$

Let $m > n \geq 0$. We apply the triangle inequality repeatedly to bound the distance between x_m and x_n :

$$d(x_m, x_n) \leq d(x_m, x_{m-1}) + d(x_{m-1}, x_{m-2}) + \dots + d(x_{n+1}, x_n)$$

Substituting the bound for consecutive terms, we obtain:

$$d(x_m, x_n) \leq (c^{m-1} + c^{m-2} + \dots + c^n) d(x_1, x_0) = c^n \sum_{k=0}^{m-n-1} c^k d(x_1, x_0)$$

Because $c \in [0, 1)$, the infinite geometric series $\sum_{k=0}^\infty c^k$ converges to $\frac{1}{1-c}$. Therefore, we can strictly bound the finite sum:

$$d(x_m, x_n) \leq c^n \frac{1}{1-c} d(x_1, x_0)$$

Let $\epsilon > 0$. Since $c < 1$, $\lim_{n \rightarrow \infty} c^n = 0$. Thus, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$:

$$c^n < \frac{\epsilon(1-c)}{d(x_1, x_0)}$$

(If $d(x_1, x_0) = 0$, the sequence is constant and trivially Cauchy). For all $m > n \geq N$, this forces $d(x_m, x_n) < \epsilon$. Thus, (x_n) is a Cauchy sequence.

Because X is a complete metric space, the Cauchy sequence (x_n) converges to a limit $x^* \in X$. We now verify that x^* is a fixed point. A contraction mapping is globally uniformly continuous (letting $\delta = \epsilon$ works for any $c < 1$). Because T is continuous, the limit commutes with the function application:

$$T(x^*) = T\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} T(x_n) = \lim_{n \rightarrow \infty} x_{n+1} = x^*$$

Hence, x^* is a fixed point of T , completing the proof.